KOSMOS

> ***K**orean **O**pen **S**ervices **M**ulti-agent **O**rchestration **S**ystem*
>
> *5,000+ API*
>
> ***Target**: KSC 2026 · **License**: Apache-2.0 · **Stack**: Python 3.12 + FriendliAI K-EXAONE + Ink TUI*

---

1. □□□ □□□□

`data.go.kr`□ □□□ 5,000□ □□ □□ API□ □□□ □□□□ □ □□ □□□ □ □□ □□□ □□□□□□□ □□□.

```
□□:   "□□ □□□□ □□ □□□, □□□ □□ □□□□"
KOSMOS: KOROAD □□ □□□ + KMA □□□□ + □□ □□ □□ □□
        → "□□□□□□ □□-□□ □□ □□ □□, □□ □□□.
           □□□□ □□□ □□□□□."

□□:   "□□□ □□ □□□ □□ □□ □□□ □□□?"
KOSMOS: 119 □□□□ API + HIRA □□□□ □□
        → □□□ □□ □□□ + □□ □□□□

□□:   "□□ □□□ □□□□ □□□"
KOSMOS: □□□□□ □□□□ API + Gov24 □□ API
        → □□ □□, □□ □□, □□□ □□ □□□
```

□□□ □□ □□□ □□ API□ □□□□□ □□□□ □□□□. □□□□ **KOSMOS□ □□□.**

1.5. □□ □□□ — □□□□ AI □□□□ 2026-2028

KOSMOS□ □□□ □□□ □□□ □□□ □□ □□□, □□ □□□ □□AX □□ □□□ □□ □□□□□. □□ □□□ □□ □□□□□□□□□□□□**,** □□□□□ □□□□ □□□□□**, 2026.2.25** □□ □□□□□□.

Flagship □□ — Open API · OpenMCP (KOSMOS □□□ 1:1)

> *"Open API OpenMCP . Open API OpenMCP ."*
>
> *— , , 2026.2.25, p.97 (7 AX, 58 — 9)*

KOSMOS□ □□ □ "□□□□ □□□□□□ □□□ □□□□ □□" □ □□ □□□□ □□□□□.

AI-Native □□ □□

> *" . AI ('AI-Native') … , ."*
>
> *— , , 2026.2.25, p.85 (7 AX)*

Citizen Scenarios □□ — □□□□·□□□·□□ID

> *" ."*
>
> *— , p.96, 1*
>
> *" , ."*
>
> *— , p.96, 5*
>
> *" AI / , ."*
>
> *— , p.96, 8*

□ □ □□□ §7 Citizen Scenarios(□□·□□·□□□□□ □)□ □□□ □□□□□. □□ □□ 5(□□ → □□□□) □ §4 Permission Pipeline □ read-only □□□□ □□□□ □□□□ □□ □□□□□□.

Permission Pipeline □□□ — AI □□□□ □□ □□

> *"AI . . ."*
>
> *— , p.14-15 (1 — AI ·)*

§4 □ **bypass-immune 7□□ □□□**, §3 □ classifier □□(prose □□□), §5 □ □□ □□ □□ □□ □□ □□ □ □□□ □□□□□.

□□□ AI □□□□ — □□ □□□

> *" AI , 'AI ' ."*
>
> *— , p.133 (11 AI)*
>
> *" . . . AI · ."*
>
> *— , p.133*

KOSMOS□ □□□ □□ □□□□□(□□ □□·□□ API □□ □□ □□)□ □ □□□ □□□ □□ □□□□□.

□□□□ □□ — □□ □□□

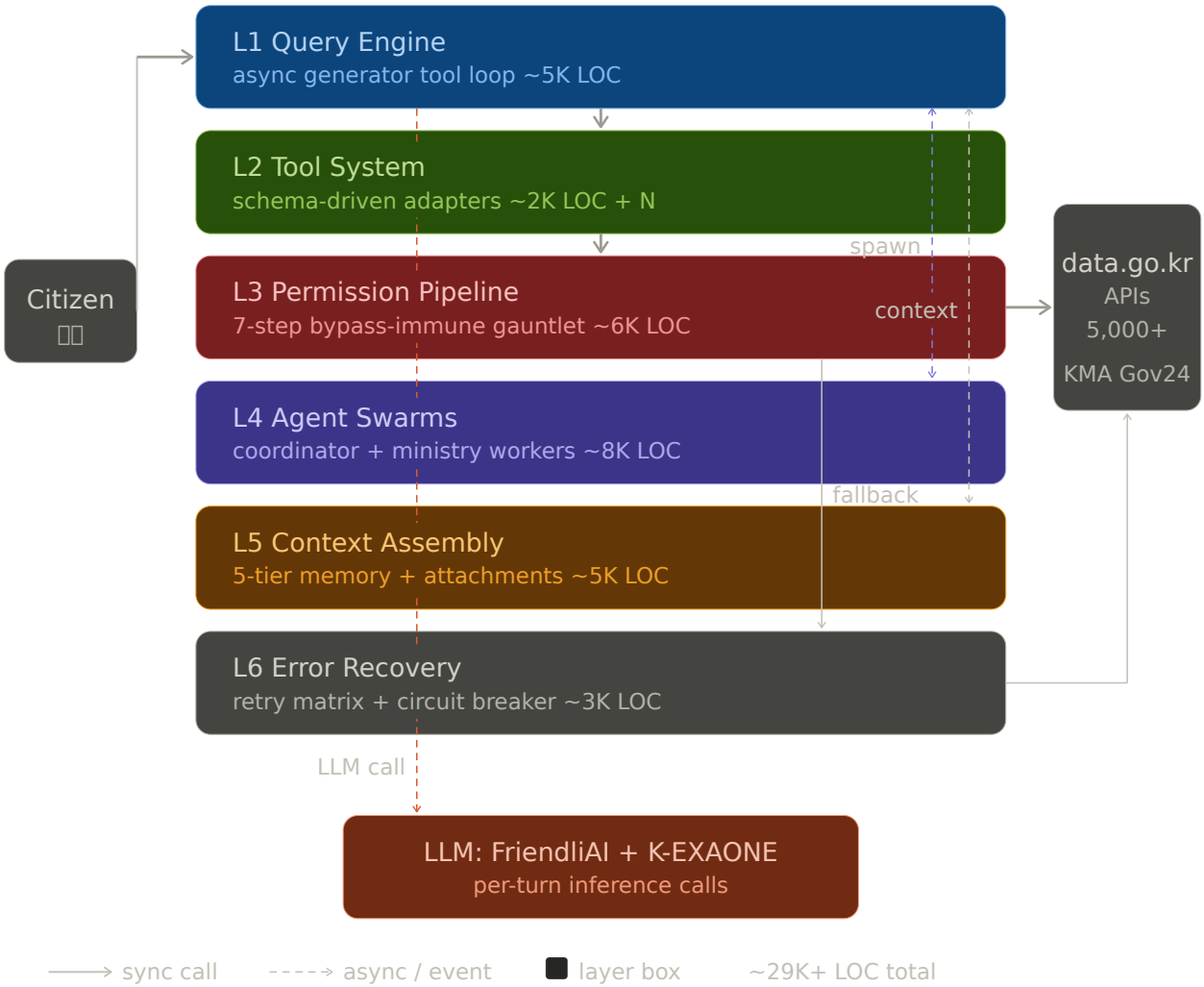
> *" AI AI ."*
>
> *— , p.97, 11*

§5 Agent Swarm □ □□ tool registry · □□ LLM □□□□□ · prompt cache partitioning □□□ □ □□□ □□□□ □□□ □□□□□.

> *: ( ), ( ), (2026.2.25), . KOSMOS , . .*

---

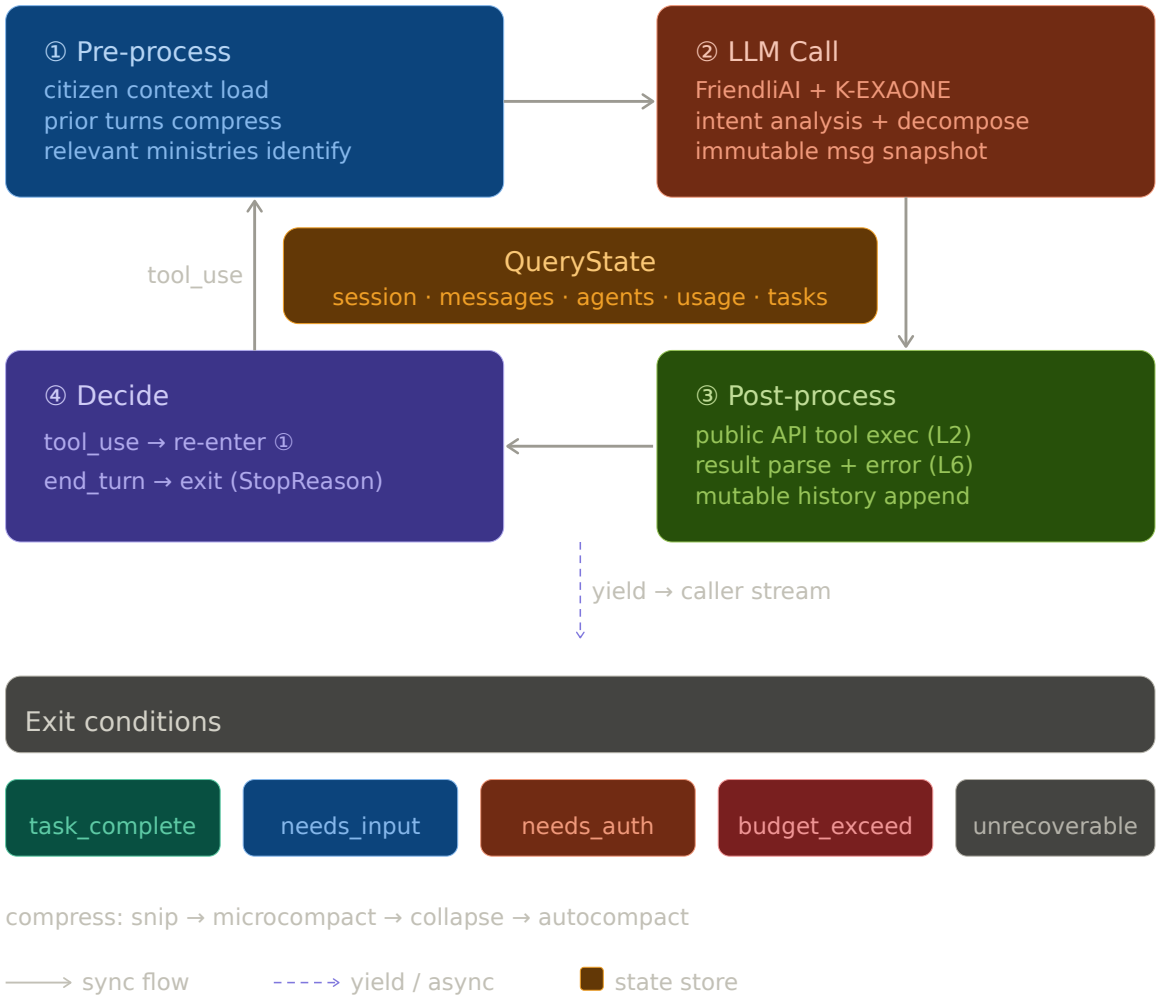
2. 6-Layer Architecture — □□ □□



□□□ **L1 Query Engine**□□ □□□ □□□ □□□, □□ □□(L2) → □□ □□(L3)□ □□ `data.go.kr` API□ □□□, L4(□□□□ □□)·L5(□□□□ □□)□ □□□□ L1□ □□□ □□□□□. **LLM**□ □ □□□ L1□ □□□□ □□ □□ □□□□□.

#	Layer	□□	□□ □□
1	**Query Engine**	□□ □□□ □□□ □□□ □□ `while(True)` □□ □□	Async generator state machine
2	**Tool System**	`data.go.kr` API □□□ □□□□□ + □□□	Schema-driven tool modules
3	**Permission Pipeline**	□□ □□ + □□□□ □□ □□□	Multi-step bypass-immune gauntlet
4	**Agent Swarms**	□□ □□ □□□□ + □□□□□	Mailbox IPC + coordinator synthesis
5	**Context Assembly**	□ □ LLM□ □□ 3□ □□□□	System + memory + attachments
6	**Error Recovery**	□□ API □□·□□·□□ □□	`withRetry` □□□□

3. Layer 1 — Query Engine □□ □□



Query Engine□ 4□□ □□ □□(Pre-process → LLM Call → Post-process → Decide)□ □□□□□. `tool_use` □ ①□ □□□, `end_turn` □□ 5□□ □□ □□ □ □□□ □□□□□□. □□□ **QueryState**□ □□·□□□·□□□□·□□□ □ □□ □□□ □□□□, `yield` □ □□□ caller□□ □□□ □□□□□□□.

□□ □□ □□ 3□□

1. **Async generator**□ □□ □□□□□ — □□·□□□ □□ □□. `yield` □ □□ □□□□ □□□ caller□ □□ □□□ □□(backpressure). □□□ "□□"□ □□□ cancellation□ □□ in-flight API □□□ □□ □□.
2. **Mutable history + immutable per-call snapshots** — □□ □□□□ in-place□ □□□□□, LLM □□□□ immutable copy □ □. **prompt cache**□ □□□ □□ □□□ □□ — □□ □□□□ cache invalidate□□ □□□ □□□.
3. **Multi-stage preprocessing** — `tool-result budget → snip → microcompact → collapse → autocompact`. □□ + □□ □□ + □□□□ □□□ □ □□□ □□□□ □□□□ □□□ □□ □□.

□□ □□ 5□□

```
StopReason:
  task_complete           # □□ □□
  needs_citizen_input     # □□ □□ □□
  needs_authentication    # □□□□ □□
  api_budget_exceeded     # □□ □□ □□
  error_unrecoverable     # □□ □□
```

---

4. Layer 3 — Permission Pipeline 7□□ □□□

□□ tool call□ □□□ □□□□ □□ 7□□ □□ □□□. □□ □ □□ □□□□ **Refusal Circuit Breaker**□ □□□□, N□ □□ □□ □ □□□/□ □ □ **Human Channel**□ □□□□□□.

#	□□□	□□ □□
1	Configuration	API□ □□ □□(public/authenticated/restricted)
2	Intent Analysis	□□□ □□□ □ □□ □□□ □□□□□□
3	Parameter Inspection	□□□ □□□ □□ □□ □□ PII□ □□□□□
4	**Citizen Authentication**	□□□□ □□□□ □□□ □□□□□□ ⛔ bypass-immune
5	Ministry Terms-of-Use	□□□ □□□ □□□□ □□ □□
6	Sandboxed Execution	□□□ □□□□□□ ambient credential □□ □□
7	Audit Log	□□·□□ID·API·□□·□□ □□ □□

Bypass-immune □□

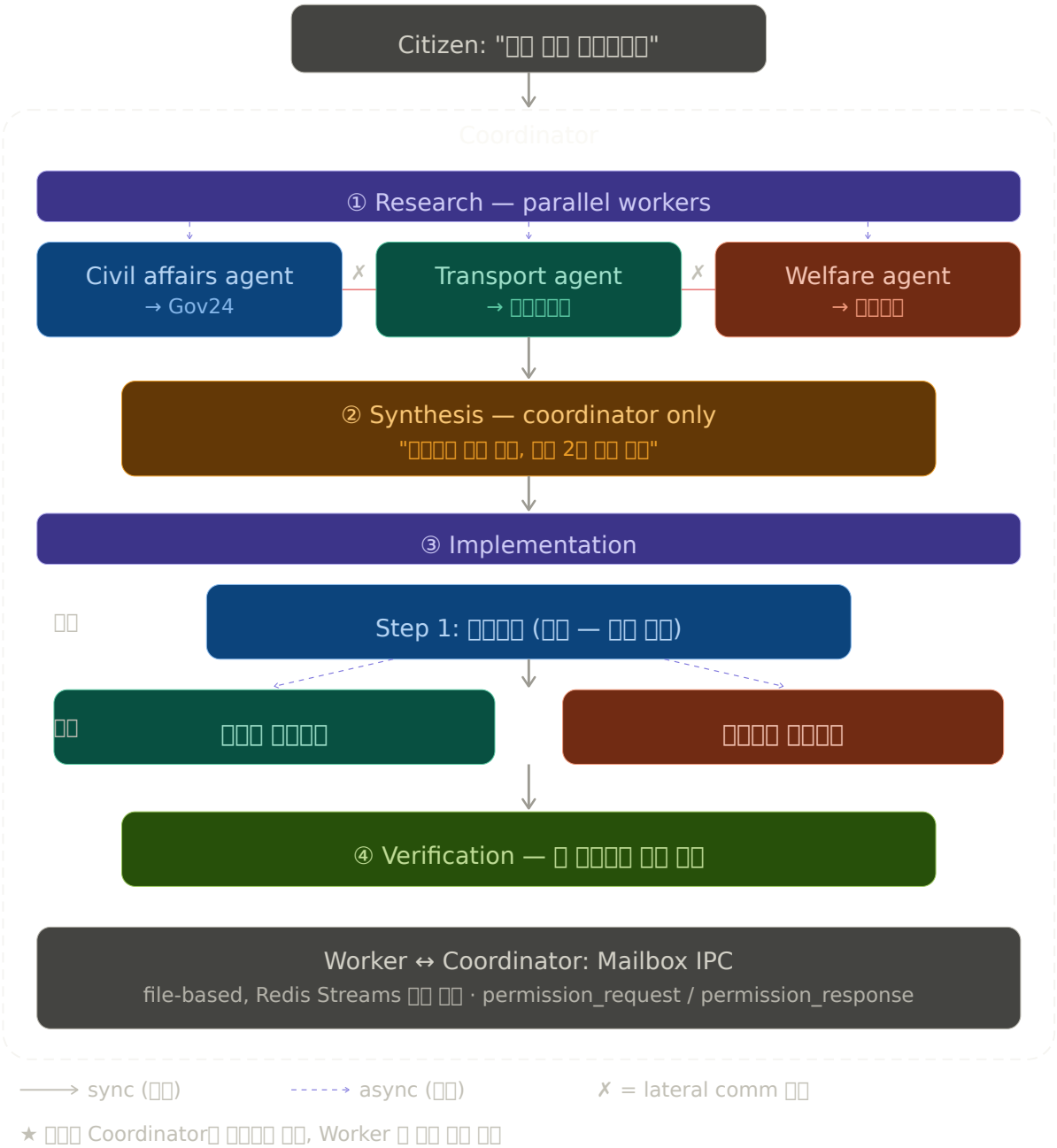
□□ □□□□ □□ □□**(□□□/□□□/YOLO)**□□ □□ □□: - □□□ □□□□ □□ - □□□ □□ □□ □□□□ □□ - □□□□ □□ □□□ □□□ □□ □□(□□·□□·□□)

Classifier □□

LLM □□ □□ □□□ □□□□ □□□ **tool call**□ □□□ □□, □□□□□ □□□ □□□ □□(prose)□ □□ □□ □□. □□□ □□□□ □□□ □□ □□□ □□ □□.

---

5. Layer 4 — Agent Swarm: □□ □□□□



□□□ □□□ □□ □□□□ □□□□□ □□□□□. KOSMOS□ □□□□□-□□ **swarm**□ □□□□□.

```
□□: "□□ □□ □□□, □□□□□ □□□ □□□□□□
      □□□□ □□□□ □ □□ □□□"

Coordinator:
  Research (□□ □□):
    ├─ □□24 □□□□  → Gov24 □□□□ □□
    ├─ □□ □□□□    → □□□ □□ □□ □□
    └─ □□ □□□□    → □□□□ □□ □□

  Synthesis (□□□□□ □□, □□ □□):
    "□ □ □□□□□ □□□□□ □.
     □□ □□□·□□□□□ □□ □□."

  Implementation:
    Step 1: □□□□ (□□ — prerequisite)
    Step 2: □□□ + □□□□ (□□ — independent)

  Verification (□□):
    └─ □ □□□□ □□ □□
```

□ □□ □□ □□

- **Coordinator owns synthesis** — □□□ raw findings□ □□. □□·□□□ □□□□□ □□.
- **Permission never flows laterally** — □□□ □□□ □□□□ `permission_request` □ □□□□□□ □□ □□□□ □□, □□ □□□ □□ □□ □□□ □□□□. □□-□□ □□ □□ □□.

Mailbox IPC

□□-□□□□□ □□□ in-process callback□ □□□ **durable mailbox**. □□□ □□ □□ mailbox□ □□□□, production□ Redis Streams□ □□□□□□. □□ □□□□□ □□.

□□: □□□□ □ □□, □□□ □□□(□□□ □□), □□□ □□□(mailbox □□ inspect), □□□□□ service discovery □□□.

---

6. Layer 6 — Error Recovery □□□□

```
Public API call → error?
  ├── 429 Rate limited     → exponential backoff (base 1s, cap 60s)
  ├── 503 Maintenance      → □□ API □□ → □□□ □□□□ □□
  ├── 401 Auth expired     → □□ □□ □ 1□ retry
  ├── Timeout              → ×3 □□□, □□ □□□ fallback
  ├── Data inconsistency   → □□ □□ API□ cross-verify
  └── Hard failure         → graceful □□□ + □□ □□□ □□
```

Foreground vs background: □□□ □□ □□ □□ foreground□ □□ retry, □□·□□□□□ □□ □□ background□ fail-fast.

---

7. Citizen Scenarios — □□ □□

□□□□ □□ 5□□□ end-to-end□ □□□□ □□□ □□ □□□:

1. **Route safety** — `KOROAD + KMA + □□ □□` → □□ □□ □□
2. **Emergency care** — `119 + HIRA` → □□ □□□ □□
3. **Childbirth benefits** — `MOHW □□ + Gov24 □□` → □□ □□ + □□ □□□
4. **Residence transfer** — `Gov24 + □□□□□ + □□□□` □□
5. **Disaster response** — `KMA + NEMA + □□□ □□` □□

---

8. KOSMOS□□ □□□

□□□□ □□ □□□□(Claude Code, Gemini CLI, Aider …)□□□ □□□□ □□ □□ □□□ □□ □□:

- **Bilingual tool discovery** — 5,000+ □□ □□□□ □□ API□□ □·□ □□ □□
- **Bypass-immune permission pipeline** — □□ PII □□ (PIPA □□, □□□ □□ □□)
- **Multi-ministry coordination** — □□ □□□ □□□□ □□□ (□: □□□□ → □□□ □□□□)
- **Prompt cache partitioning** — □□□□ □□□□ □□ AI □□ □□□
- **Fail-closed adapters** — □□□ □□□□ deny, allow □□

---

9. □□□ □□ — 2026 LLM-Agent Production-Grade

Phase 1(CLI MVP) □□ □□ **Infra Initiative**(GitHub #462) 6□ Epic □□ □□:

Epic	□□	□□ □□
#463	**Observability**	OpenTelemetry GenAI semconv + Langfuse self-host
#464	**Evals & Regression**	Promptfoo PR gate + DeepEval pytest-style
#465	**Cost & LLM Gateway**	LiteLLM Proxy + per-session budget
#466	**Safety Rails**	Presidio PII + Guardrails AI + indirect-injection □□
#467	**CI/CD & Prompt Registry**	uv Docker + prompt manifest + shadow/canary
#468	**Secrets & Config**	Infisical OIDC + 12-factor cleanup

□□: OWASP LLM Top 10 2026, "Bypassing Guardrails" arXiv 2504.11168, Snyk 2025□ GitHub □□□□ □□ 28M□ □□□.

---

10. Roadmap

- **Phase 1 — Prototype** ✅ — FriendliAI Serverless + 10□ □□□ API + □□ query engine + CLI. Scenario 1 end-to-end □□ (live 33/33 pass)
- **Phase 2 — Swarm** — □□ □□ □□□□, mailbox IPC, □□ API □□. Scenarios 1–3
- **Phase 3 — Production** — □□ □□ □□□□□, □□□□, □□ □□, □ □□□□, □□ □□

Code Scope Estimate

Layer	LOC
Query Engine	~5,000
Tool System	~2,000 + N adapters
Permission Pipeline	~6,000
Agent Swarms	~8,000
Context Assembly	~5,000
Error Recovery	~3,000
Total core	**~30,000**

---

11. References

□□□□ □□ □□□□·□□·□□□ □□□ □□ □□:

□□	□□□□	□□
Claude Agent SDK	MIT	Async generator tool loop, permission types
OpenAI Agents SDK	MIT	Guardrail pipeline, retry matrix
Pydantic AI	MIT	Schema-driven tool registry, graph state machine
AutoGen (Microsoft)	MIT	AgentRuntime mailbox IPC, InterventionHandler
Anthropic Cookbook	MIT	Orchestrator-workers □□
Gemini CLI (Google)	Apache-2.0	Ink + React + Yoga TUI □□ □□
Claude Code sourcemap	Reconstructed	□□ □□ □□, □□ □□, □□□□ □□
PublicDataReader	MIT	`data.go.kr` □□□ □□ ground truth
NeMo Guardrails (NVIDIA)	Apache-2.0	Colang 2.0 — PIPA □□ □□□ □□ □□
Google ADK	Apache-2.0	Runner-level plugin, reflect-and-retry
LangGraph	MIT	`RetryPolicy`, `ToolNode(handle_tool_errors=True)`
stamina / aiobreaker	MIT	Async retry + circuit breaker (L6)
arXiv 2601.06007	Open access	"Don't Break the Cache" — 30–50+ tool-call□□ 41–80% □□ □□

□□ □□: `docs/vision.md § Reference materials`

12. Non-goals — □□□ □□□

- □□ □□ □□□□□ □□ — □□ □□·□ □□ □□ □□
- □□ □□ □□□□ □□ — □□ □□□□ □□□ □, □□ □□ □□ □□
- □□ **API** □□ □□□ □□ — □□ □□□ 6□ □□□□ □□ □□

> ***Source of truth***: `docs/vision.md` · ***Operational rules***: `AGENTS.md` ***License***: *Apache-2.0* · ***Repo***: *github.com/umyunsang/KOSMOS*